MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

§5.2

Addition and Subtraction with Integers

Number lines, charged chips, and the definitions

Today's Goal

After completing this section, you should be able to:

- Add two integers using the number line model, drawing a directed arrow for each step
- Add two integers using the charged chip model, recognizing that a positive chip and a negative chip combine to make zero
- State the definition of integer addition, in both its case-by-case form and its absolute value form, and use either one to evaluate a sum

Today's Goal, Continued

- Subtract two integers using the number line model, turning the person around to change which direction is forward
- Subtract two integers using the charged chip model, adding pairs of chips that make zero when there are not enough chips to remove
- Rewrite a subtraction as addition of the opposite, and use that to evaluate a difference

Learning Target 13

LT.13 Arithmetic with Integers

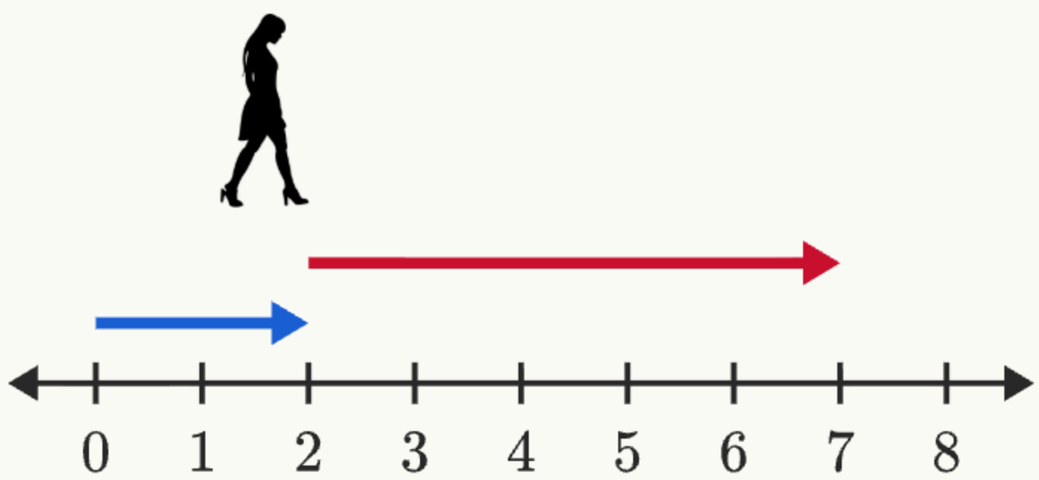
I can apply multiple strategies (including the standard algorithms) for addition, subtraction, multiplication, and division of integers using appropriate representations.

Number Line Model for Addition

When using the number line model for addition with integers, we can use a person walking along the number line to help us. For example, we can represent $2 + 5$ on the number line in two stages.

1. We begin with 2, which is a positive number. So on the number line model, the person starts at 0, faces in the positive direction, and moves forward 2 units.

2. Then we add 5 by moving forward another 5 units.



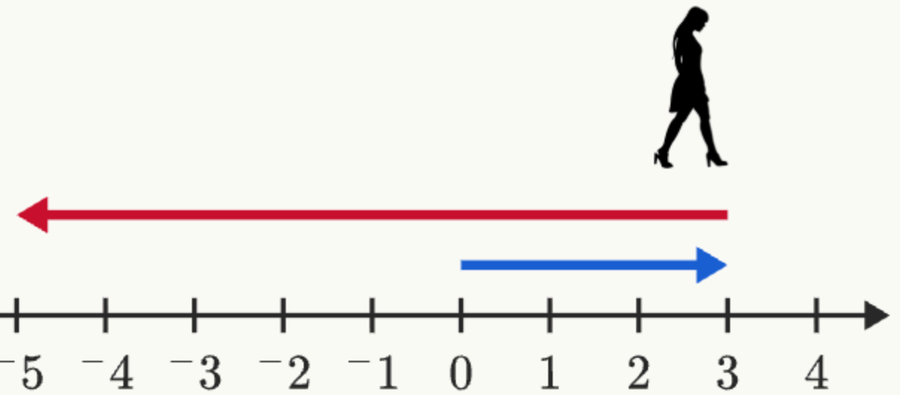
This shows that $2 + 5 = 7$.

Adding on the Number Line

Evaluate $3 + (^-8)$ using the number line model.

1. We begin with 3, which is a positive number. So on the number line model, the person starts at 0, faces in the positive direction, and moves forward 3 units.

2. Then we add $^-8$. Since we are adding, we continue facing forward. But we are adding $^-8$, so we move backwards 8 units.

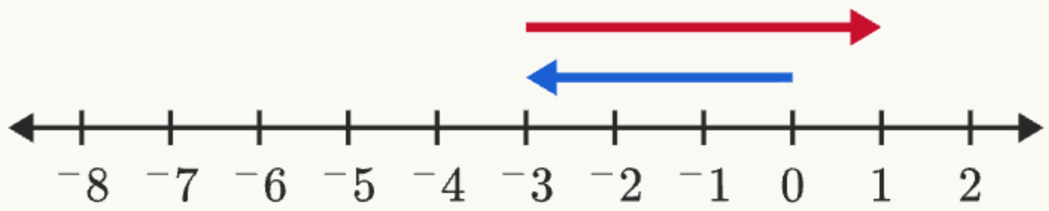


This shows that $3 + ^-8 = ^-5$.

What Went Wrong?

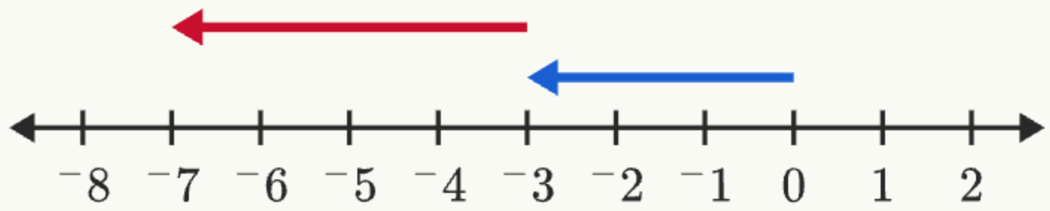
A student used directed arrows on the number line to evaluate $^{-}3 + {}^{-}4$ and got 1. What mistake did the student make?

The student's work:



$^{-}3 + {}^{-}4 = 1$

The correct work:

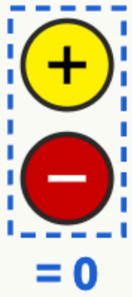


$^{-}3 + {}^{-}4 = {}^{-}7$

The student moved forward 4 units. Since we are adding, we continue facing forward. But we are adding $^{-}4$, so we move backwards 4 units.

Charged Chip Model for Addition

When using the charged chip model for addition, note that one positive chip and one negative chip will always combine to make zero because $1 + {}^{-}1 = 0$.

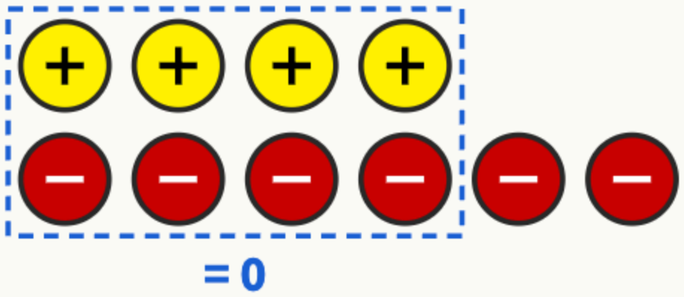


In fact, we can generalize this and say that n positive chips and n negative chips will always make zero because $n + {}^{-}n = 0$.

Adding with Charged Chips

Use the charged chip model to represent and evaluate $4 + {}^{-}6$.

We need 4 positive (yellow) chips and add 6 negative (red) chips.



Since we can match up 4 positive chips with 4 of the negative chips, those will combine to make zero.

This leaves us with 2 negative chips. So $4 + {}^{-}6 = {}^{-}2$.

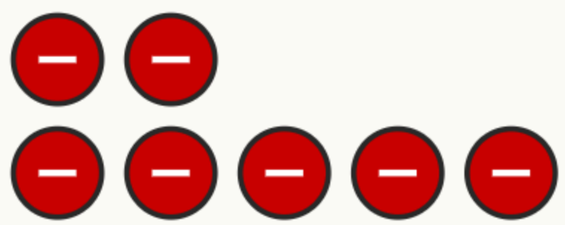
Same Signs

Determine whether or not the following statements are true.

a. When we add two positive integers, the result is always a positive integer.

b. When we add two negative integers, the result is always a negative integer.

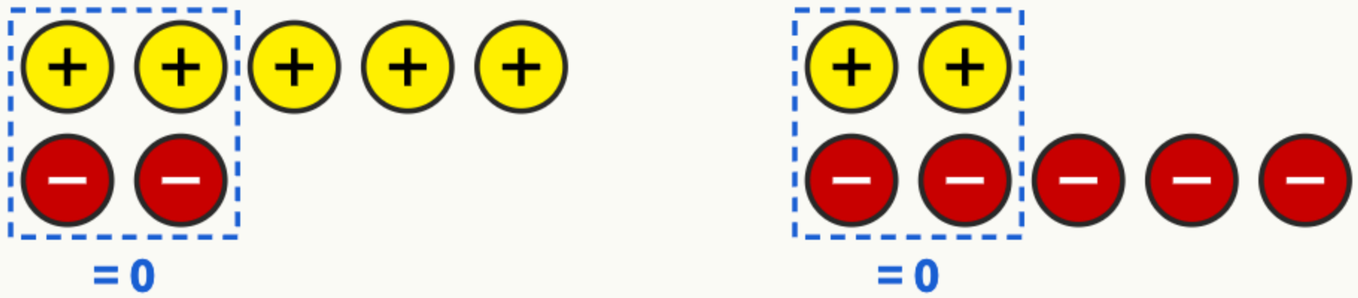
Try one: $^-2 + ^-5$. No chip combines to make zero, so $^-2 + ^-5 = ^-7$.



Both are true. The two numbers have the same sign, so no chip is ever canceled.

What Decides the Sign?

Compare $5 + {}^{-}2$ and $2 + {}^{-}5$. What decides whether the answer is positive or negative?



$$5 + {}^{-}2 = 3 \qquad 2 + {}^{-}5 = {}^{-}3$$

The sign of the answer depends on which kind of chip runs out first. How many chips are left is the difference $5 - 2 = 3$.

Defining Addition with Integers

Definition 5.2.5

Let m and n be any two integers.

a. $m + 0 = 0 + m = m$

b. *(Adding Two Non-negative Integers)* If $m \geq 0$ and $n \geq 0$, then $m + n$ is defined using addition of whole numbers.

c. *(Adding Two Negative Integers)* If $m > 0$ and $n > 0$, then $^{-}m + {}^{-}n = {}^{-}(m + n)$.

Defining Addition, Continued

Definition 5.2.5, continued

d. *(Adding a Positive Integer and a Negative Integer)*

If $n > 0$ and $m \geq n$, then $m + {}^{-}n = {}^{-}n + m = m - n$.

If $n > m$ and $m > 0$, then $m + {}^{-}n = {}^{-}n + m = {}^{-}(n - m)$.

Addition with Absolute Values

Definition 5.2.9

Let m and n be any two integers.

a. $m + 0 = 0 + m = m$

b. *(Adding Two Non-negative Integers)* If $m \geq 0$ and $n \geq 0$, then $m + n$ is defined using addition of whole numbers.

c. *(Adding Two Negative Integers)* If $m < 0$ and $n < 0$, then $m + n = {}^{-}(|m| + |n|)$.

Absolute Value Form, Continued

Definition 5.2.9, continued

d. *(Adding a Positive Integer and a Negative Integer)*

If $m > 0, n < 0$, and $|m| \geq |n|$, then $m + n = n + m = |m| - |n|$.

If $m > 0, n < 0$, and $|m| < |n|$, then $m + n = n + m = {}^{-}(|n| - |m|)$.

Adding with the Definitions

Use Definition 5.2.5 and Definition 5.2.9 to evaluate each of the following.

(a) $^{-}7 + {}^{-}6$

Using Definition 5.2.5:

$$\begin{aligned} {}^{-}7 + {}^{-}6 &= {}^{-}(7+6) \\ &= {}^{-}13 \end{aligned}$$

Using Definition 5.2.9:

$$\begin{aligned} {}^{-}7 + {}^{-}6 &= {}^{-}(|{}^{-}7| + |{}^{-}6|) \\ &= {}^{-}(7+6) \\ &= {}^{-}13 \end{aligned}$$

(b) $9 + {}^{-}4$

Using Definition 5.2.5:

$$\begin{aligned} 9 + {}^{-}4 &= 9 - 4 \\ &= 5 \end{aligned}$$

Using Definition 5.2.9:

$$\begin{aligned} 9 + {}^{-}4 &= |9| - |{}^{-}4| \\ &= 9 - 4 \\ &= 5 \end{aligned}$$

(c) $4 + {}^{-}11$

Using Definition 5.2.5:

$$\begin{aligned} 4 + {}^{-}11 &= {}^{-}(11-4) \\ &= {}^{-}7 \end{aligned}$$

Using Definition 5.2.9:

$$\begin{aligned} 4 + {}^{-}11 &= {}^{-}(|{}^{-}11| - |4|) \\ &= {}^{-}(11-4) \\ &= {}^{-}7 \end{aligned}$$

Number Line Model for Subtraction

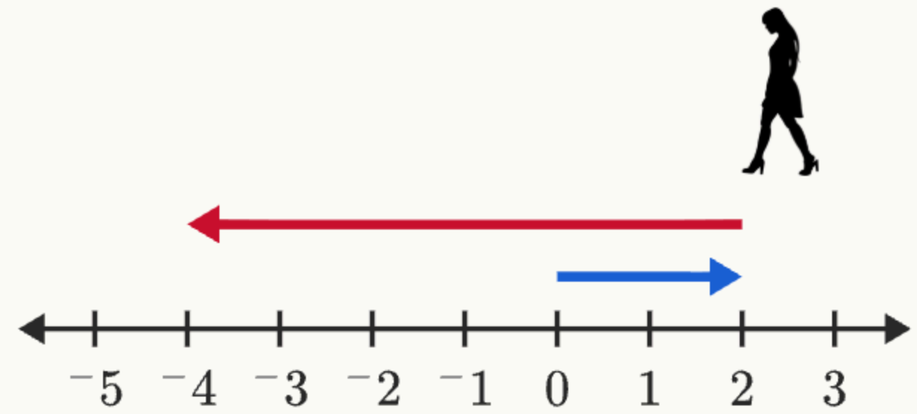
When using the number line model for subtraction with integers, we also use a person walking along the number line to help us. However, when subtracting, the person turns around, which changes which way is forward and backward.

For example, we can represent $2 - 6$ on the number line in two stages.

Representing $2 - 6$

1. We begin with 2, which is a positive number. So the person starts at 0, faces in the positive direction, and moves forward 2 units.

2. Then we subtract 6. Since we are subtracting, the person turns around to face in the negative direction. This makes the negative direction forward and the positive direction backward. 6 means to move forward 6 units.

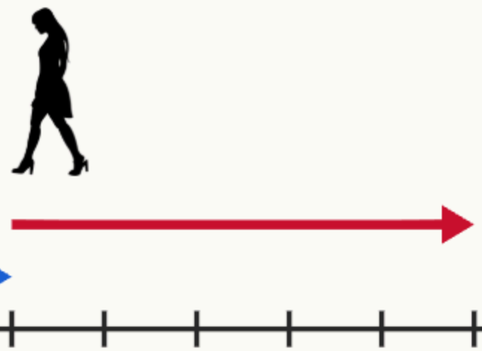


This shows that $2 - 6 = {}^{-}4$.

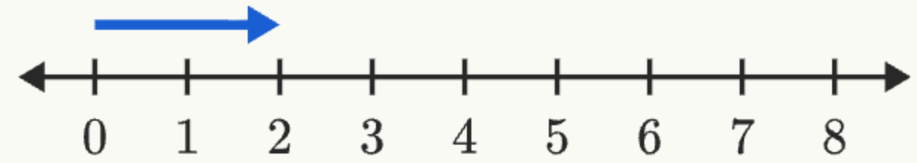
Evaluate $2 - {}^{-}5$

Evaluate $2 - {}^{-}5$ using the number line model.

1. We begin with 2. The person starts at 0, faces in the positive direction, and moves forward 2 units.



2. Then we subtract ${}^{-}5$. Since we are subtracting, the person turns around to face in the negative direction. This makes the negative direction forward and the positive direction backward. ${}^{-}5$ means to move backward 5 units.



This shows that $2 - {}^{-}5 = 7$.

Charged Chip Model for Subtraction

Using the charged chip model, remember that “subtracting” means to remove chips.

In particular, “subtracting a positive integer” means to remove positive chips and “subtracting a negative integer” means to remove negative chips.

Subtracting with Charged Chips

Use the charged chip model to represent and evaluate $^{-}7 - {}^{-}3$.

We need 7 negative (red) chips to start with.



Subtracting $^{-}3$ means to remove 3 negative (red) chips.



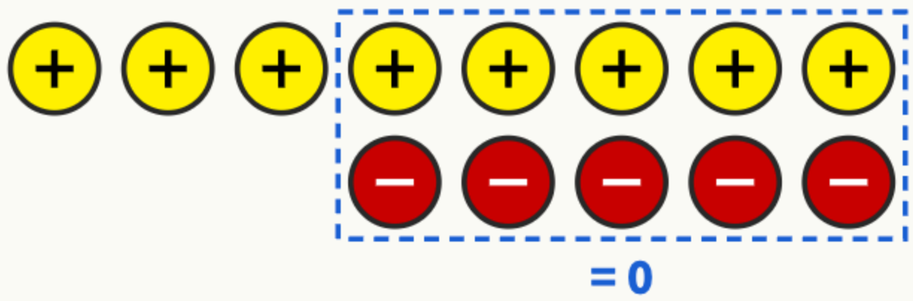
This leaves us with 4 negative (red) chips. So $^{-}7 - {}^{-}3 = {}^{-}4$.

Not Enough Chips

How can we use charged chips to evaluate $3 - 8$? We have to remove 8 positive (yellow) chips, but we only have 3.



We address this by adding 5 pairs of positive and negative chips.

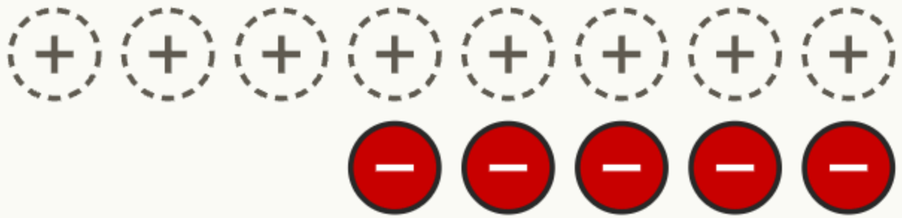


Not Enough Chips, Continued

Why is it allowed to add those pairs?

These pairs of positive and negative chips we added in are equivalent to zero, so we are not changing the value being represented. The chips still represent the number 3.

Now we have enough positive chips so that we can remove 8 of them.



This leaves us with 5 negative (red) chips. So $3 - 8 = {}^{-}5$.

Subtracting a Negative

Use the charged chip model to explain why $4 - {}^{-}6$ is equivalent to $4 + 6$.

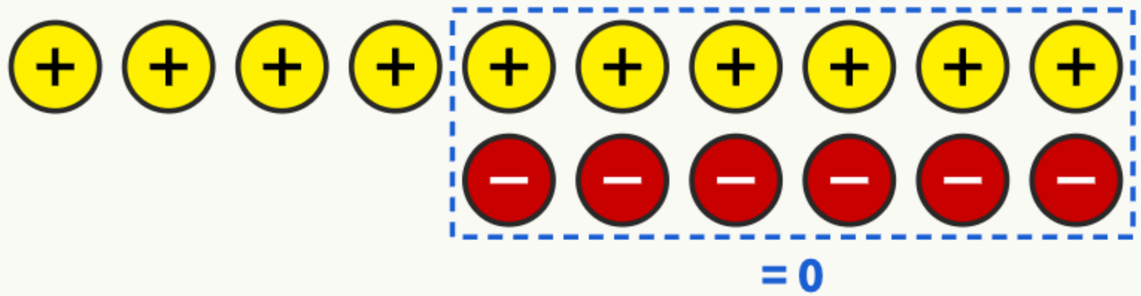
We need 4 positive (yellow) chips to start with.



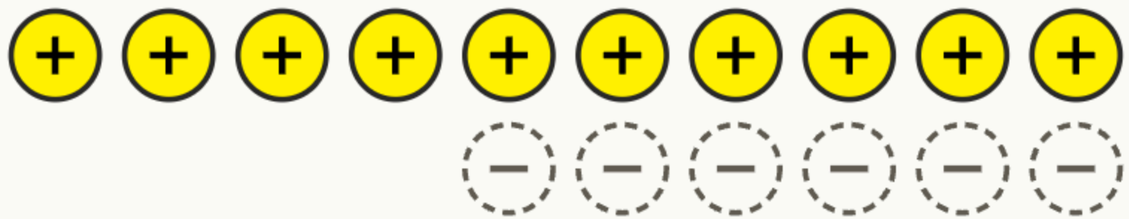
Subtracting ${}^{-}6$ means to remove 6 negative (red) chips. This is a problem because there are no negative chips at all.

Subtracting a Negative, Continued

We address this by adding 6 pairs of positive and negative chips.

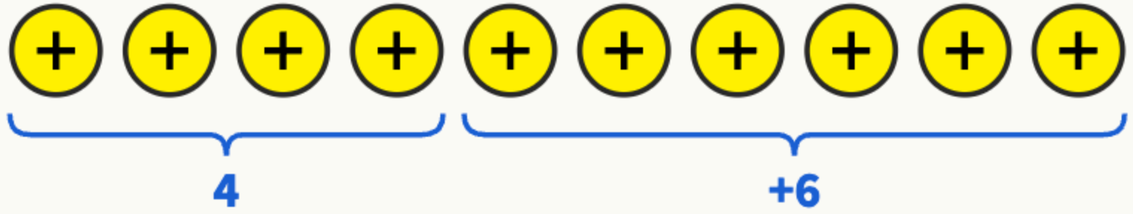


The chips still represent 4. Now we have enough negative chips so that we can remove 6 of them.



Subtracting a Negative, Continued

The 6 negative (red) chips were only ever there as partners for 6 positive (yellow) chips. Taking the negative chips away leaves those 6 positive chips behind with nothing to cancel them.



So subtracting 6 negative chips had exactly the same effect as adding 6 positive chips, and $4 - {}^{-}6 = 4 + 6 = 10$.

Defining Subtraction with Integers

Definition 5.2.16

If m and n are any two integers, $m - n$ is the unique integer d such that $m = n + d$.

For example, suppose that we want to evaluate $9 - 13$. Let $d = 9 - 13$. We need to determine the value of d.

By Definition 5.2.16, d is the unique integer such that $9 = 13 + d$.

Since we can check that $13 + {}^{-}4 = 9$, this means that $d = {}^{-}4$.

Subtraction as Adding the Opposite

The result from Example 6 is a specific example of the more general fact that subtracting a number can be thought of as adding the opposite of that number.

Theorem 5.2.17

If m and n are any two integers, then

$$m - n = m + {}^{-}n.$$

Using the Theorem

Use Theorem 5.2.17 to evaluate each of the following expressions.

(a) $^{-}36 - 18$

We interpret "subtracting 18" as "adding the opposite of 18".

$$\begin{aligned} ^{-}36 - 18 &= {}^{-}36 + {}^{-}18 \\ &= {}^{-}54 \end{aligned}$$

(b) $24 - {}^{-}15$

We interpret "subtracting $^{-}15$" as "adding the opposite of $^{-}15$".

$$\begin{aligned} 24 - {}^{-}15 &= 24 + {}^{-}({}^{-}15) \\ &= 24 + 15 \\ &= 39 \end{aligned}$$

Subtract by Adding the Opposite

Use Theorem 5.2.17 to evaluate each of the following expressions.

(a) $28 - 53$

$$\begin{aligned} 28 - 53 &= 28 + {}^{-}53 \\ &= {}^{-}(53 - 28) \\ &= {}^{-}25 \end{aligned}$$

(b) ${}^{-}41 - {}^{-}29$

$$\begin{aligned} {}^{-}41 - {}^{-}29 &= {}^{-}41 + 29 \\ &= {}^{-}(41 - 29) \\ &= {}^{-}12 \end{aligned}$$